

A gammapy-plugin for threeML

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<https://github.com/threeML>

The Multi-Mission Maximum Likelihood¹ Framework

(Vianello, Lauer, et al., 2015)



Also called threeML or 3ML

¹"Maximum Likelihood" for historical reasons

What is threeML?

The Goal

Fit data from multiple instruments jointly with as little sacrifices as possible

The Idea

- Use the official software for anything instrument related
- Provide a common structure for the instruments (the PluginPrototype)
- Provide a modeling package - `astromodels` (Vianello, Burgess, et al., 2021)
- Have one shared model being fitted/sampled and support many samplers & minimizers

3ML Workflow

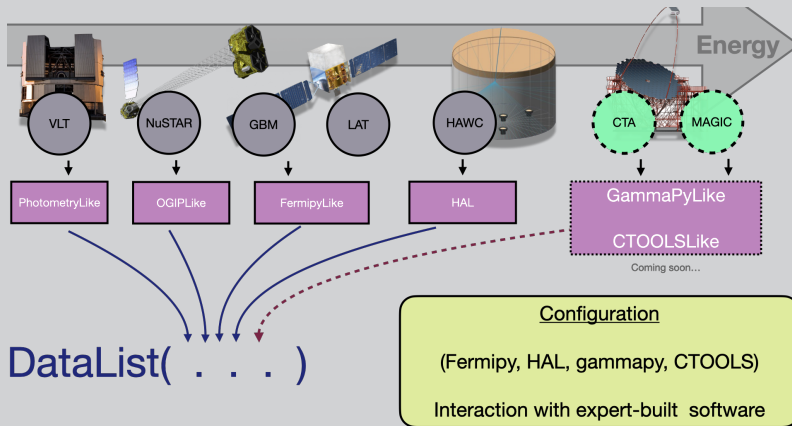


Figure: Source: Internal threeML resource

3ML Workflow

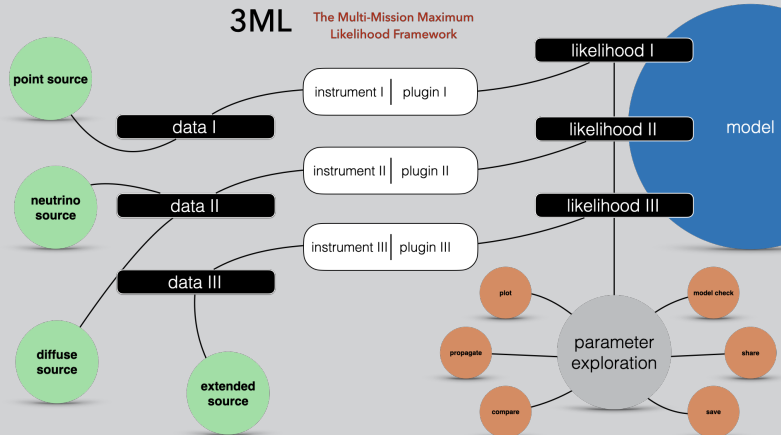
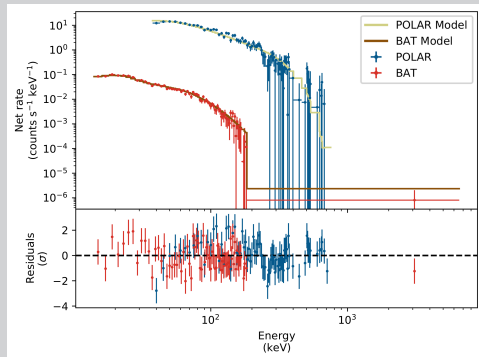


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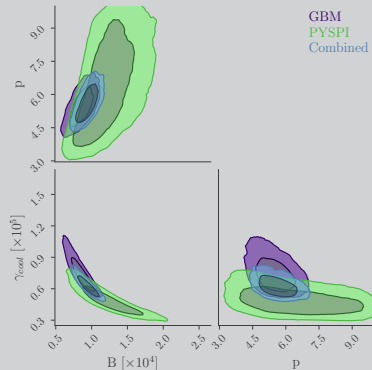
What's supported by threeML?

- Any OGIP dataset (XRT, BAT, XMM-Newton, GBM, ... via OGIPLike)
- Fermi/LAT (via FermipyLike - uses fermipy)
- Fermi/LAT (via FermiLATLike - uses fermitools) → Unbinned Likelihood
- COSI (via cosipy once launched)
- INTEGRAL/SPI (via pyspi - soon not just for transients)
- Swift/BAT (via BatAnalysis soon)
- POLAR and POLAR2 (via polpy soon)
- IXPE (via ixpepy)
- HAWC (via HAL)
- IceCube (via i3mla)
- Optical data (via PhotometryLike)
- ...
- Your gammapy dataset (via gammapy-plugin soon)

3ML MWL Examples



Joint Swift/BAT + POLAR Spectrum & Polarization
Fit for GRB 170101A (Kole et al., 2020)



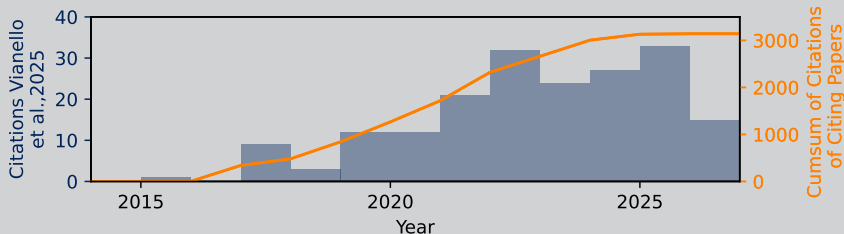
Joint Fermi/GBM + INTEGRAL/SPI Synchrotron Fit
for GRB 120711A1 (Biltzinger et al., 2022)

Why do I bore you with this?



Why do I bore you with this?

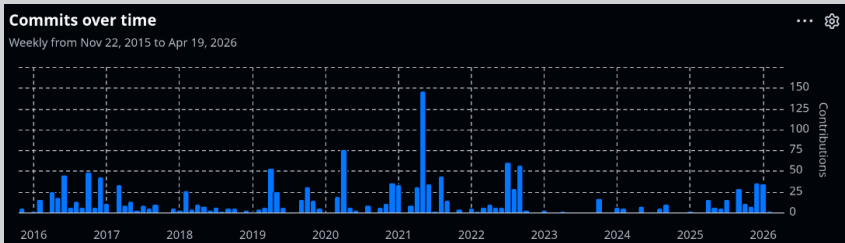
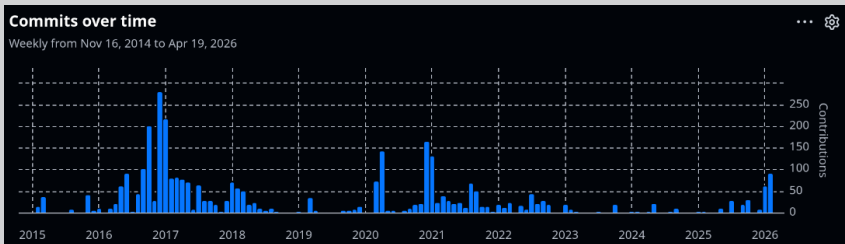
- Created a `gammapy`-plugin for 3ML 🤔
- Will make methodically & statistically sound MWL analysis easier 🎉
 - Best way of handling the data right out of the box
 - If instrument-specific plugin exists → no limitations by common data format
 - User-friendly interface to samplers and results (fully written in Python 🐍)
 - Proven and tested MWL framework:



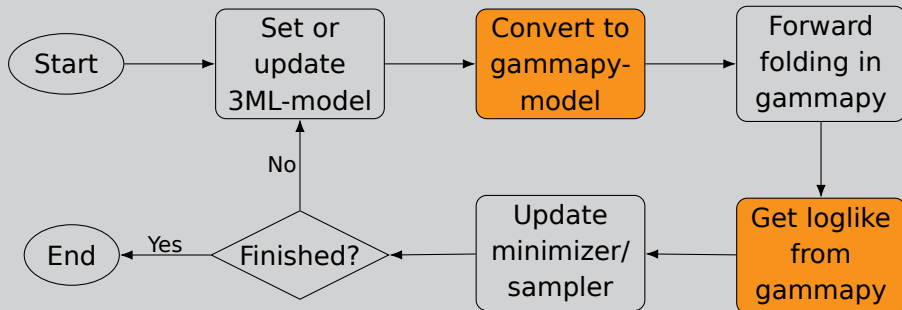
Why go the "hard" way

- Flux Points are only valid for the model they have been computed for
- Combining multiple instruments with different angular/energy resolution can be challenging and systematic uncertainties can be introduced
- Really hard to deal with upper limits

We are not dead (yet)



The Basic Concept of the `gammapy`-plugin



What can currently be done

- Only fitting spectra 
- 2D and 3D without some bodgy unit transformations 
- Including FOVBackgroundModels 
- If you can model it in astromodels → we can fit it 

Disclaimer

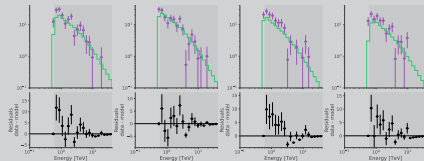
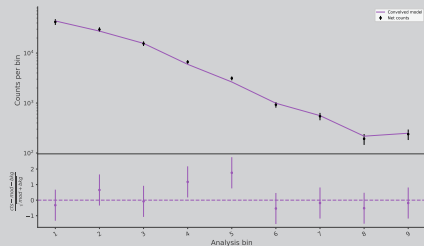
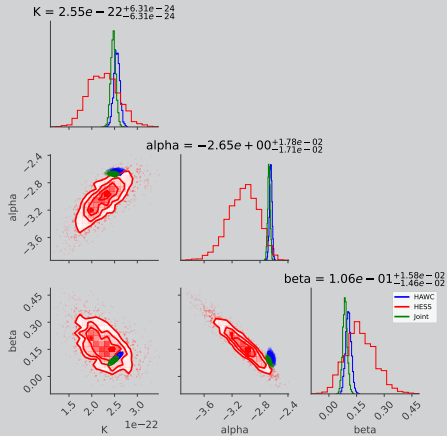
The following are not analysis results!

Just proofs of concepts

Exclusively public datasets

MWL 1D-Example - H.E.S.S. + HAWC via HAL

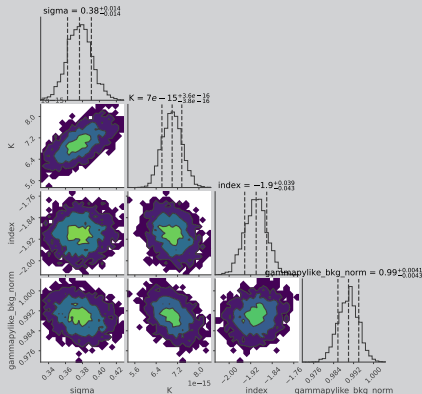
HAWC example borrowed from Henrike Fleischhack² Data from Abeysekara et al., 2017



²<https://github.com/henrikef/threeML-analysis-workshop/>

3D-Example

RX J1713.7-3946 with a Gaussian Morphology (yes, that's very wrong) + Powerlaw Spectrum



Morphology:

$$\sigma_r \approx 0.38^\circ \checkmark$$

Background Model:

$$\mathcal{N}_{\text{bkg}} \approx 0.99 \checkmark$$

Spectrum:

$$7 \times 10^{-15} \text{ TeV} - 1\text{cm} - 2\text{s} - 1 \text{ 😞}$$

missing factor of $\left(\frac{180}{\pi}\right)^2 \dots$

$$\Rightarrow 2.3 \times 10^{-11} \text{ TeV} - 1\text{cm} - 2\text{s} - 1 \checkmark$$

Roadmap

- ① Fix unit and frame issues within 3ML and `astromodels`
- ② Improve plotting + visualizing analysis results
- ③ Continue validation of the `gammapy`-plugin
- ④ Release to public
- ⑤ Add "reverse" converter → convert `gammapy` models to `astromodels`

Thank you!

Thank you for your attention! I

-  Abeysekara, A. U. et al. (June 2017). “Observation of the Crab Nebula with the HAWC Gamma-Ray Observatory”. In: *The Astrophysical Journal* 843.1, p. 39. ISSN: 1538-4357. DOI: 10.3847/1538-4357/aa7555.
-  Biltzinger, Björn et al. (July 2022). “Improving INTEGRAL/SPI Data Analysis of GRBs”. In: *Astronomy & Astrophysics* 663, A102. ISSN: 0004-6361, 1432-0746. DOI: 10.1051/0004-6361/202243189.
-  Kole, M. et al. (Dec. 2020). “The POLAR Gamma-Ray Burst Polarization Catalog”. In: *Astronomy & Astrophysics* 644, A124. ISSN: 0004-6361, 1432-0746. DOI: 10.1051/0004-6361/202037915.
-  Vianello, Giacomo, J. Michael Burgess, et al. (Nov. 5, 2021). *Astromodels*. Zenodo. DOI: 10.5281/zenodo.5646925.

Thank you for your attention! II

 Vianello, Giacomo, Robert J. Lauer, et al. (July 2015). “The Multi-Mission Maximum Likelihood Framework (3ML)”. In: arXiv:1507.08343. DOI: 10.48550/arXiv.1507.08343. arXiv: 1507.08343 [astro-ph].

Backup Slides

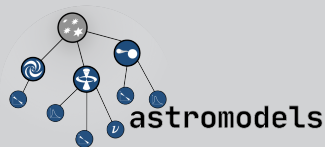
Workflow

- 1 Create your `gammapy` dataset as usual
- 2 Design your model using `astromodels`
- 3 Wrap your dataset in a `GammapyLike` plugin
 - 1 When multiple datasets: decide if one plugin per dataset or one plugin for multiple ones
 - 2 Option for also setting a background model per plugin via a `.set_background_model()` function
- 4 Add the Instance to your `3ML DataList` (container of all the plugins)
- 5 Create an Analysis Instance (links your Model and DataList)
- 6 Setup the sampler and start it
- 7 Grab at least one coffee

What about modelling?

astromodels

- Uses fixed units under the hood → reduced overhead due to astropy units
- ronswanson for building table models
- Support for XSPEC functions
- Also easily expandable (see e.g. pynchrotron)



Who might benefit

- Are you working with MWL data?
- Do you have access to the data of the instruments?
- You do not want to make any compromises on data formats?
- Can you model it in `astromodels`?
- You do not need all analysis/plotting capabilities of `gammapy` (yet)?
- You want to try multiple samplers/minimizers?